

ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

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ABSTRACT

The question of when and why adversarial examples (AExs) materialise in neural networks (NNs) remains fundamentally unresolved, with prevailing accounts construing them either as artefacts of irregularities in the decision landscape or as manifestations of a heightened sensitivity to non-robust input features. In contradistinction to these positions, we advance the thesis that adversarial vulnerability can originate in the *efficiency* with which NNs encode information. Concretely, we demonstrate that superposition – the regime in which networks represent a greater number of features than they possess dimensions – engenders configurations of latent representations that adversaries are able to exploit. Our analysis establishes that adversarial perturbations capitalise on the interference between superposed features, rendering attack patterns predictable from the underlying feature arrangements. The resulting framework furnishes a mechanistic account of two hitherto empirically documented phenomena: the transferability of adversarial attacks between models trained under comparable regimes, and the emergence of class-specific vulnerability patterns. Within synthetic settings that afford precise control over the degree of superposition, we establish that superposition *suffices* to induce adversarial vulnerability. We subsequently demonstrate that these findings persist in a vision transformer (ViT) trained on CIFAR-10. Taken together, these results reveal that adversarial vulnerability may constitute a byproduct of the representational compression networks employ, rather than a symptom of deficiencies in the learning process or of non-robust inputs. We release our implementation through a software repository.

1 INTRODUCTION

Notwithstanding the considerable body of research devoted to AExs (Szegedy et al., 2014; Goodfellow et al., 2014), no consensus has crystallised regarding their fundamental origin. A complete explanation of how minimal perturbations can so drastically reconfigure model predictions continues to elude the field, leaving us unable to anticipate which perturbations will succeed, to account for the transfer of attacks across models, or to formulate principled defences.

Extant explanations coalesce broadly into two camps (Nakkiran, 2019). The *bug* perspective ascribes vulnerability to a range of factors, among them deficiencies in learning (Schmidt et al., 2018; Nakkiran, 2019) and the geometric properties of high-dimensional decision boundaries (Fawzi et al., 2016). Work in this tradition concentrates on establishing statistical bounds on quantities such as capacity (Bubeck & Sellke, 2021) or Lipschitz constraints (Hein & Andriushchenko, 2017), yet stops short of explaining which specific perturbations succeed. The *feature* perspective, by contrast, contends that AExs exploit predictive but non-robust statistical regularities in the data (Ilyas et al., 2019), thereby recasting the debate in terms of the relationship between the underlying data and human perception. In so doing, it treats these non-robust features as fixed properties of the data rather than as consequences of the encoding strategies a network adopts. Neither account, however, reconciles the manner in which representational constraints interact with data semantics.

In the present work, we bridge this divide by adopting a mechanistic interpretability perspective, demonstrating that adversarial vulnerability arises from the interplay between architectural constraints and data semantics (*i.e.* the human-interpretable properties of the data). We show that AExs can emerge from interference among the learned representations of NNs. Adversarial attacks systematically exploit superposed latent structure – a mechanism that ordinarily furnishes additional representational capacity – in order to construct effective perturbations that manipulate model out-

puts. Our analysis discloses a mechanistic pathway along which input correlations constrain feature arrangements, these arrangements in turn determine interference patterns, and these interference patterns ultimately dictate both attack characteristics and transferability. The resulting framework renders tractable the prediction of which perturbations succeed and why they transfer between models.

Our account of AExs is grounded in recent theories concerning how networks encode and process information, namely the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH posits that *input features*—the fundamental abstractions of data—are encoded as linear directions within a network’s representation space. It is further hypothesised that, through superposition, NNs can represent substantially more such features than they possess neurons, thereby packing multiple features efficiently into shared dimensions at the cost of introducing interference. Such interference entails that perturbing one feature may affect others in non-obvious ways. This paper interrogates whether the interference so induced gives rise to vulnerabilities that AExs exploit, and what insight this affords for the understanding of adversarial phenomena. Our principal contributions are as follows:

- We advance a framework that reconciles architectural constraints with feature properties in the aetiology of adversarial vulnerability. Employing simple models, we demonstrate how data properties induce specific superposition geometries that, in turn, directly determine the realised adversarial perturbations. We further show that interference patterns illuminate both attack transferability and class-specific susceptibility.
- We establish that these findings persist in a ViT trained on CIFAR-10, augmented with an engineered bottleneck layer that induces superposition.
- We demonstrate that superposition is sufficient, though not necessary, for adversarial vulnerability. In doing so, we identify algorithmic brittleness as a distinct vulnerability mechanism.
- We show that a mechanistic understanding of learned representations enables the informed construction of adversarial attacks, thereby underpinning our argument for semantically-informed defences.

2 BACKGROUND

We now furnish an overview of the conceptual apparatus underpinning our analysis, namely the LRH and superposition. Formally, let $\mathbf{x} \in \mathcal{X}$ denote the input to a NN and $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ the activation vector of the l -th layer.

Linear representation hypothesis (LRH). The LRH posits that NNs encode many of the variables of their computation—among them the semantic properties of their inputs—as linear directions within their activation space, directions that may thereafter serve as abstractions for reasoning (Park et al., 2024; Guerner et al., 2023). This predisposition toward linear feature representations is hypothesised to arise because linear separability enables networks to recognise and manipulate features with ease, and because the dot products taken against subsequent-layer weights process such directional features efficiently. A growing body of evidence corroborates this view (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (e.g. concepts such as “presence of shape” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions $\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that differ mainly in the value of feature c_j while holding other features approximately constant (i.e., $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k \mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs are capable of representing considerably more latent features than there are dimensions available in activation space: $M > d_l$. LLMs, for instance, can invoke far more place names than their residual stream affords dimensions. One framework for analysing this phenomenon is superposition, according to which networks deploy an overcomplete and non-orthogonal set of feature directions $\{\mathbf{v}_j\}_{j=1}^M$. This strategy capitalises on two facts: that $2^{\Theta(d\epsilon^2)}$ almost-orthogonal vectors (of $< \epsilon$ cosine similarity) can be accommodated within a d -dimensional space (Tkocz, 2012), and that sparse vectors admit accurate recovery following projection into lower-dimensional spaces (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This economy, however, incurs two attendant challenges. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) emerges, whereby individual neurons contribute to multiple distinct features, so that a neuron’s activation no longer corresponds to any single interpretable concept. Second, non-orthogonal feature directions engender *interference* among features—the activation of one feature partially activates others. Networks can attenuate these difficulties through non-linear operations (e.g. ReLU, softmax) that serve to disambiguate superposed features (Gurnee et al., 2023).

Definition 2 (Superposition Hypothesis) *A network layer represents features in superposition if:*

1. *The number of latent features exceeds layer dimensionality: $M > d_l$*
2. *There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$*
3. *Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$*

In packing more features than it has dimensions, the network trades augmented representational capacity against the feature interference thereby incurred.

Adversarial attacks. We next delineate the notions of adversarial perturbation and attack, following (Szegedy et al., 2014; Costa et al., 2024):

Definition 3 (Adversarial attack) *Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p -sense, e.g. $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small δ that p -perturbs $\mathbf{x}$ to create an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ (untargeted) or $f(\mathbf{x}') = y_{\text{target}}$ (targeted) is known to be an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are created via the projected gradient descent (PGD) (detailed in Sec. A).*

Definition 4 (Adversarial robustness) *A classifier $g_w(\cdot)$ with parameters w is said to be **robust** at $\mathbf{x}$ with radius $\epsilon > 0$, if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.*

3 SUPERPOSITION GEOMETRY CAN DETERMINE ADVERSARIAL ATTACKS

We begin by interrogating whether, and by what means, adversarial attacks exploit the interference rooted in the superposition of latent features. Specifically, we pose three questions:

1. *Do adversarial perturbations exploit the interference between superposed features?*
2. *Do correlations in the input shape the geometric arrangement of superposed latent features?*
3. *Can shared latent geometry explain why attacks transfer between independently trained models?*

Experimental setup. In order to address these questions, we devise a synthetic task that instantiates established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022) within a precisely controllable setting. The task is engineered so as to: (1) afford an intuitive classification setting for the study of AExs; (2) encode class concepts as linear directions in accordance with the LRH; (3) induce controlled superposition among these latent features; and (4) preserve a priori knowledge of the correspondence between inputs and superposed features — thereby enabling testable predictions concerning adversarial mechanisms.

Concretely, we partition the input $\mathbf{x} \in \mathbb{R}^d$ into k equally-sized groups $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, wherein $\mathbf{x}^{(j)}$ denotes the p input features associated with class j . The task consists in identifying the group of largest sum: $y = \operatorname{argmax}_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$. Intuitively, each $x_i^{(j)}$ constitutes evidence for class j , the p features together capturing distinct class attributes. Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ with sparsity S (the probability of setting an entry to zero).

A two-layer network compresses the input into m dimensions by means of a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (where $m < k < d$), and subsequently reconstructs it through a

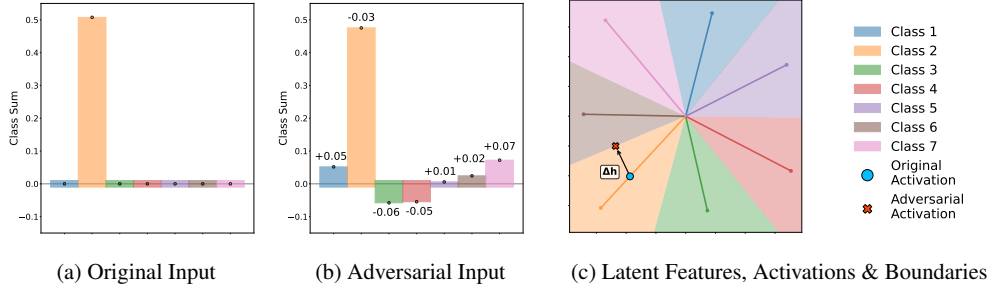


Figure 1: **An adversarial attack exploiting superposition geometry** ($k = 7, m = 2$). (a) The original sample. (b) The adversarially perturbed sample, whose ground truth is preserved yet which is nonetheless misclassified; the sign and magnitude of each input perturbation are dictated by the configuration of the latent representations. (c) The original and adversarial samples situated in activation space, the arrows denoting the column vectors of $\mathbf{W}_e$, that is, the latent representations of the input features.

decoder, $\mathbf{z} = \mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary configuration adopts cross entropy (CE) loss in the absence of ReLUs and biases, a choice motivated by its simplicity, the softmax serving to curtail interference by exponentially suppressing the contributions of interfering latent features. To guard against the possibility that our findings are peculiar to this configuration, we corroborate them within a second setup employing mean squared error (MSE) loss together with ReLUs and biases. The representational challenge, then, is to compress information from $d = k \times p$ dimensions into $m < k$ dimensions and thereafter recover sufficient information to identify the class of largest sum.

Whereas non-synthetic tasks would furnish features that encode nuanced concepts requiring discovery and interpretation, the present setup supplies ground-truth class labels for the latent features. This provision permits us to compute the true class following any perturbation simply by recalculating group sums, which in turn facilitates our definition of an AEx as a perturbation satisfying the following two conditions:

1. The model’s prediction under perturbation changes (*e.g.* from class A to B)
2. The true class remains unchanged (*e.g.* still class A)

This criterion serves to distinguish genuine AExs from perturbations that in fact alter the ground truth (*i.e.* that change which group attains the highest sum). Attacks are constructed using ℓ_∞ - and ℓ_2 -norm PGD.

Evaluation metrics. We construe the columns of the encoder weight matrix $\mathbf{W}_e$ as the latent representations $\mathbf{v}_i$ corresponding to each input feature x_i . Since input feature $x_i^{(j)}$ contributes exclusively to class j , these $\mathbf{v}_i$ (for $i \in \text{class } j$) become mutually aligned. This direct correspondence between input features and class-aligned latent directions enables us to trace with precision how perturbations to the inputs of a given class, $\mathbf{x}^{(j)}$, propagate into changes in latent space. For successful PGD-generated AExs $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we measure the following:

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? Although an attack must displace a sample across a decision boundary in order to alter its class, the question of what determines the requisite input perturbations remains. Two contrasting intuitions present themselves: the *feature intuition*, which anticipates an increase in the input features of the target class, and the *bug intuition*, which holds that adversarial perturbations modify inputs in unpredictable ways, such that knowledge of feature representations affords no insight whatsoever.

Fig. 1 exhibits a representative AEx within our setting, depicting an input of Class 2 (orange) alongside its perturbed counterpart (annotated with its IPP), which is misclassified as Class 6 (brown). The perturbations present an arbitrary appearance—we manifestly do not simply increase the target-class features, in contradiction of the feature intuition. Yet neither are they random. Rather, they conform to a precise pattern exhibiting a systematic correspondence between δ and the configuration of the latent features. This relationship intimates a third intuition: that *attacks are mediated by latent geometry*.

Quantitatively, the IPP exhibits a pronounced correlation with the latent attack alignment. The sign and magnitude of the perturbations $\delta^{(j)}$ correlate strongly with the degree to which their class’s latent representation $\mathbf{v}^{(j)}$ aligns with the latent attack vector $\Delta\mathbf{h}$. Positively aligned representations ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) undergo a proportional amplification of their input features, whereas negatively aligned representations suffer attenuation. It is precisely this behaviour that Fig. 1(b) illustrates—input perturbations dictated by the dot products of Fig. 1(c) so as to displace the sample across the decision boundary.

To ascertain whether PGD attacks specifically exploit superposition geometry, we juxtapose PGD-generated attacks against the theoretically optimal perturbations that we show to leverage superposition (derived in Sec. 3.2). For each configuration (k, m) , we train 5 models under distinct random seeds and generate 1000 PGD attacks per model, computing the cosine similarity between each successful perturbation and its corresponding optimum. We further establish a random baseline by generating perturbations of matching ℓ_2 norm and measuring their similarity to the theoretical optimum. Tab. 1 attests to a near-perfect alignment between PGD and optimal attacks across a range of dimensionalities. To ensure that only meaningful perturbations are retained, we exclude samples whose ℓ_2 norm falls below ϵ . One-sample t-tests comparing the observed similarities against the random baseline yield $p < 10^{-10}$ for every configuration, demonstrating that PGD attacks systematically accord with the theoretical predictions rather than arising by chance. Given that optimal attacks leverage interference and that PGD attacks attain near-perfect alignment with these optima, we conclude that PGD exploits superposition interference.

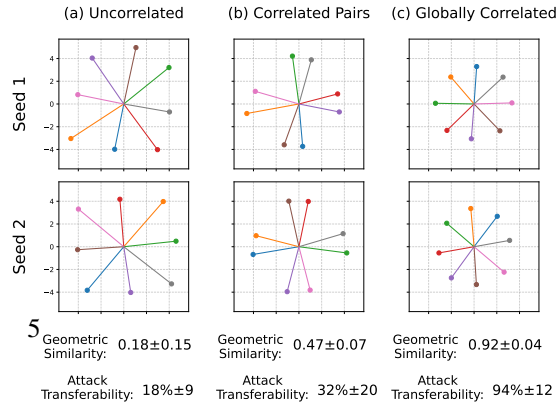
Finding: Adversarial attacks systematically exploit the interference between superposed features. Successful PGD attacks, far from being random, are predictable given the specific superposition geometry.

Do input correlations determine latent feature geometry? We turn now to the question of whether correlations within the input data determine the geometric arrangements of the latent features. By systematically manipulating the correlation structure of the training data, we probe how such correlations constrain the degrees of freedom available to the learned representations. To this end, we introduce three correlation patterns of graduated constraint strength. Under the *uncorrelated* condition, features are sampled independently, as in the preceding subsection. Under *paired* correlations, each feature in class i is coupled with a feature in class j , such that the activation of one entails the activation of its pair. Under *global* correlations, each sample is assigned a random phase that governs a cyclic activation pattern: classes adjacent in index possess similar activation probabilities and thus co-activate frequently, whereas classes more widely separated seldom activate in concert.

Fig. 2 portrays how these correlations govern the arrangement of $\mathbf{v}^{(j)}$. We quantify geometric similarity by comparing the pairwise cosine-similarity matrices computed over all feature pairs within each model, and thereafter measuring the correlation between these matrices across distinct random seeds; a higher correlation is indicative of more closely matching geometric arrangements.

Table 1: Alignment between PGD-discovered attacks and the theoretically optimal perturbations across configurations. Results are reported as mean $\pm$ std over 1000 attacks per condition, with all p -values falling below 10^{-10} .

k	m	Cosine Sim. (PGD vs Theory)	Cosine Sim. (Random Baseline)
6	2	0.97 ± 0.02	0.00 ± 0.02
30	10	0.96 ± 0.00	0.00 ± 0.01
90	30	0.92 ± 0.00	0.00 ± 0.00



For each correlation condition, we examined 25 seed pairs. In the $m = 2, k = 7$ setup, geometric similarity amounted to 0.23 ± 0.14 (uncorrelated), 0.47 ± 0.32 (paired), and 0.88 ± 0.06 (global) (mean $\pm$ std). The results evince a monotonic relationship: uncorrelated data yields geometries that vary considerably across seeds, paired correlations partially constrain the arrangements, and global correlations compel near-identical geometries. Statistical significance was assessed by pairwise two-sample t-tests among all three modes with Bonferroni correction ($p < 10^{-3}$ for all comparisons, $\alpha = 0.05$ corrected for 3 comparisons). Results spanning a range of (k, m) configurations are provided in Sec. C.3.

The operative mechanism is one of *interference avoidance*: features that co-activate frequently are arranged so as to minimise their mutual interference. Stronger correlations impose correspondingly tighter constraints—uncorrelated data admits any arrangement that separates the classes, paired correlations fix the relative positions of coupled features, and global correlations determine a unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. Stronger correlations diminish the geometric degrees of freedom, compelling different initialisations to converge upon similar arrangements.

Does shared geometry explain attack transferability? Having established that input correlations induce consistent latent geometries across model initialisations, we now examine whether the resulting similarity of interference patterns accounts for attack transferability. Should attacks indeed be tailored to specific interference patterns, they ought to transfer successfully only between models exhibiting similar geometric arrangements.

Employing the same seed pairs as in the previous paragraph, we generate adversarial attacks on source models and evaluate their efficacy when transferred to target models within each correlation condition (25 transfer measurements per condition). All models attain $> 95\%$ clean accuracy. Transfer rates correlate strongly with geometric similarity: for the $k = 7, m = 2$ configuration (Fig. 2), globally correlated data yields $94\% \pm 12\%$ attack transfer (mean $\pm$ std), uncorrelated data affords a mere $18\% \pm 9\%$ transfer, and paired correlations produce an intermediate $32\% \pm 20\%$ transfer.

Each perturbation component amplifies or suppresses features, thereby producing the constructive interference that propels representations across decision boundaries. When the identical perturbation is subsequently applied to a model of differing feature arrangement, those features that formerly interfered constructively instead cancel or interfere destructively, whereupon the attack fails.

Finding: Attack transferability is governed by shared interference patterns.

Does reducing superposition suppress these attacks? If superposition begets vulnerability by way of interference, then the removal of superposition ought to curtail adversarial vulnerability. We subject this proposition to three experiments, each of which constrains interference. In the first, we set $m = k$, whereupon the network learns to represent each class direction $\mathbf{v}^{(j)}$ orthogonally. Under this configuration, any perturbation that alters the model’s prediction necessarily alters the ground-truth class as well: to move a sample from class A to class B requires rendering the sum of the class-B features greater than that of class A—that is, genuinely transforming it into a class-B sample. We record zero successful adversarial examples across 1000 attempts at every ϵ tested. In the second, we disentangle the effect of superposition from that of capacity by fixing m while varying k . Robust accuracy decreases monotonically with the superposition pressure k/m (App. Tab. 5), demonstrating that

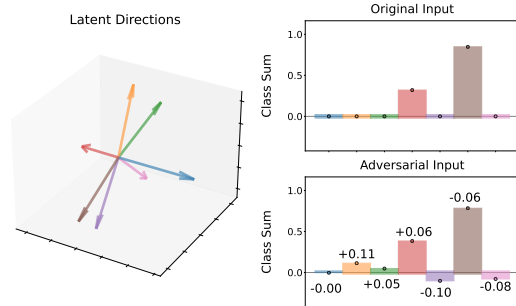


Figure 3: An adversarial attack (directed from class 5 to class 3) leaves unperturbed the input features of a class represented orthogonally (class 1).

vulnerability scales with the degree of interference. In the third, we test whether attacks concentrate upon superposed features by constraining one class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others throughout training. When attacks are generated between classes that remain in superposition (e.g. from class a to b where $\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs corresponding to the orthogonal class are left unperturbed (Fig. 3).

Finding: Adversarial attacks expend their budget preferentially upon those features held in superposition.

3.2 FORMAL ANALYSIS

We now render formal the mechanisms that underlie our empirical observations. Within our linear setting, exact relationships between superposition interference and adversarial vulnerability may be derived, thereby extending our results beyond mere observational correlation. We retain the notation of our experimental setup, in which $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition through its columns $\mathbf{v}_i$, and in which the decoder corresponds to the transpose of the encoder (as observed empirically). The logit for class j is given by $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in Sec. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This result lays bare the manner in which superposition engenders adversarial vulnerability. Each input feature i is perturbed in proportion to its interference with the class representations. Under superposition, the attendant non-orthogonality entails that semantically unrelated features nonetheless bear upon the class decision—and adversarial perturbations exploit precisely these dependencies in order to manipulate the outputs.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q} \mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in Sec. B. $\square$

In conjunction with our empirical findings, these propositions establish a **mechanistic pathway**. Specifically: (i) input correlations constrain the arrangement of features in superposition; (ii) these geometric arrangements determine the interference patterns; (iii) the interference patterns, in turn, dictate the optimal perturbations via $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1); and (iv) shared geometric constraints yield similar interference patterns across models, thereby enabling transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

4 ATTACKS IN REALISTIC MODELS

We now extend our analysis to more realistic models, specifically a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009) (together with a ResNet-18 (He et al., 2016), Sec. D.2.1). By interposing a bottleneck layer immediately prior to the final classification output, we once again compel the latent class features into a superposed representation. We thereupon examine: (i) the manner in which the degree of superposition bears upon attack transferability and robustness; (ii) whether a consistent arrangement of the superposed representations emerges across initialisations; and (iii) whether the attacks exploit feature interference.

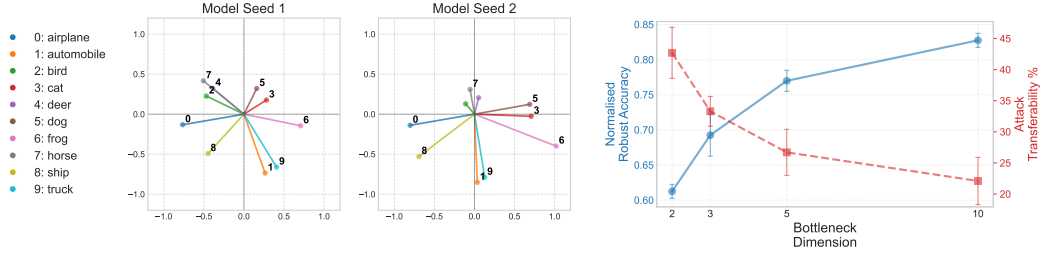


Figure 4: Left: the structure of the CIFAR-10 class representations remains similar across models trained under different seeds. Right: attack transferability and robust accuracy as the bottleneck dimension is progressively increased.

Setup. We train our ViT architecture (6 layers, patch size 4, residual stream $d = 512$) on CIFAR-10 across five random seeds. Following training, we supplant the original classification head with a bottleneck comprising a linear encoder that compresses the pre-classification activations into an m -dimensional space, succeeded by a linear decoder that expands them once more to the $k = 10$ CIFAR-10 classes. This bottleneck structure, trained with CE loss and no ReLU (thereby mirroring the setup of Sec. 3), forces the k class representations into m -dimensional superposition. Throughout training of the bottleneck layer, the weights of the base ViT are held frozen. We vary the bottleneck dimension $m \in \{2, 3, 5, 10\}$ so as to modulate the degree of superpositional compression. AExs are generated for each bottlenecked model using both ℓ_∞ - and ℓ_2 -norm PGD. We adopt a small step size in conjunction with a large number of iterations, terminating once an attack successfully alters the model’s prediction (full details in Sec. D.1). Robust accuracy is normalised by the model’s clean accuracy so as to account for the performance variations attributable to differing bottleneck capacities. Attack transferability is measured as defined in Sec. 3.

Results. The mean clean accuracy of the base models stands at 81%. As anticipated, the clean accuracy of the bottlenecked models diminishes with compression: 54%, 67%, 76%, and 81% for $m = \{2, 3, 5, 10\}$, respectively.

We advance three observations that recapitulate our toy-model results, thereby attesting to their transfer into this more realistic setting. First, Fig. 4 (right) demonstrates that, as the bottleneck m decreases (*i.e.* as superposition pressure mounts): (i) normalised robust accuracy declines, and (ii) attack transferability across differing model initialisations rises. The decline in robust accuracy under intensifying superposition is plausibly attributable to the denser packing of the class representations, which renders it easier for small input perturbations to induce widespread interference across these compressed features. The heightened transferability, in turn, follows from our Sec. 3 finding that greater superposition curtails the degrees of freedom in the potential feature geometry. With a more constrained representational space at its disposal, the network commands fewer viable geometric arrangements for its class features. This circumstance drives different model initialisations to converge upon more similar representational geometries and, in consequence, upon more thoroughly shared interference patterns, which manifest as greater attack transferability. We refer the reader to Sec. D.1.1 for results across perturbation magnitudes.

In corroboration of this, Fig. 4 (left) reveals that the geometric arrangement of the m -dimensional class representations within the bottleneck exhibits consistent ordering and clustering across models. This consistency, in accordance with the findings of Sec. 3, intimates that inherent correlations guide the formation of a superposed geometry. Notably, these groupings frequently reflect semantic affinities among the classes (*e.g.* the ‘cat’ and ‘dog’ representations clustering together). Finally, when AExs generated for a bottlenecked model are propagated through the corresponding base ViT, they occasion similar relative logit changes (*i.e.* $\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$). This concordance implies that both models respond consistently to the same perturbations, and that the perturbations are not merely an artefact of the bottleneck; in the absence of the heightened compression, however, they do not precipitate misclassifications.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLINESS

The preceding sections argued that feature superposition can *suffice* for adversarial vulnerability. Here we interrogate whether it is likewise *necessary*, by studying a setting in which features are represented

orthogonally rather than in superposition, yet remain susceptible to attack. This investigation lends support to a broader thesis developed throughout the paper: that the analysis of feature representations may be brought to bear upon robustness analysis. We pose three questions: *by what mechanism do such attacks operate; can this understanding be marshalled to construct attacks directly, without recourse to gradients; and does naive robustness certification alter the underlying failure mode, or does it merely inflate the attack budgets required?*

Setup. We train a linear encoder together with an MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$. As detailed in Sec. E and Nanda et al. (2023), the model discovers a trigonometric algorithm by which to solve this task. As one component of this learned algorithm¹ the encoder projects inputs onto non-basis-aligned linear representations that encode $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at the key frequencies $\omega_k = 2\pi k/P$. Ultimately, the MLP composes $\cos((a + b)\omega_k)$ by way of sum-angle identities, thereby attaining 100% accuracy. Crucially, these features are represented *orthogonally* (i.e. not in superposition). We generate AExs by relaxing the one-hot constraint and perturbing only the input a with PGD, thereby displacing inputs off the training manifold.

Attacks exploit the learned key frequencies. A Fourier analysis of the successful perturbations δ (App. Fig. 6) reveals that their dominant frequencies coincide precisely with the key frequencies ω_k of the learned algorithm. None (0%) of the attacks transfer to models possessing different ω_k , such as those trained from different seeds. The frequency-aligned perturbations occasion misclassification by effectively altering the sinusoidal values read off from the encoder features. The perturbation δ transforms the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$; the contributions introduced by the now non-zero inputs corrupt the sinusoidal values, thereby disrupting the MLP’s subsequent computations and precipitating misclassification. Since the model’s algorithm relies upon the constructive interference of each ω_k at the logits, the disruption of any single ω_k does not suffice; successful attacks must instead identify a δ that maximally perturbs multiple frequency components at once. A detailed account of the mechanism is deferred to Sec. E.1.

Analytic construction of attacks. Exploiting this mechanism, we are able to construct gradient-free, *informed* perturbations composed of composite sinusoidal waves of matching ω_k . On $0 + 0 \bmod P$, these attacks succeed with $\|\delta\|_2 = 0.22$, a value close to that of PGD (0.17, or $\approx 1.3\times$ the PGD budget) and far beneath the naive alternatives (single-index 0.97, ℓ_∞ noise 2.04, uniform 6.20). This furnishes empirical evidence that we are indeed attacking the correct mechanism, and demonstrates that a representation-aware conception of vulnerability can yield concrete attack recipes rather than merely post-hoc rationalisations.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). For fixed attack budgets, the *robust* accuracy improves from pre-robustified $\rightarrow$ post-robustified as follows: 0% $\rightarrow$ 85% at $\epsilon_\infty = 5e-3$, 0% $\rightarrow$ 96% at $2e-3$, and 76% $\rightarrow$ 100% at $1e-3$. Yet when subjected to PGD, the robustified model continues to fail through the very same frequency-based brittleness, albeit at the cost of larger perturbation norms. Certification thus elevates the requisite budgets without redressing the specific fragility itself, thereby motivating defences that explicitly target the identified semantics (here, frequency manipulation) rather than relying solely upon broad norm bounds.

Relation to the paper’s thesis. This section advances two points that dovetail with the paper’s broader narrative. First, superposition is *sufficient but not necessary*: orthogonal representations may nonetheless remain vulnerable to attack. Second, the appraisal of AExs through the lens of *architecture–semantics interaction* is actionable, in that it can inform both the design of attacks (by way of representation-informed perturbations) and the development of defences.

Though less central to the present argument, we further remark upon a shared dependence on sparsity assumptions: upstream violations of sparsity (e.g. noise or superposed features) can deliver effectively ‘pre-perturbed’ inputs to downstream orthogonal circuits, thereby activating the same frequency-based failure.

¹We refer to the specific computation the network executes in solving the task as the *model’s learned algorithm*. Because this algorithm depends upon the extraction of precise values from the learned feature representations (frequencies), it remains *brittle* in the face of small perturbations to the bases—a phenomenon we term *algorithmic brittleness*.

6 RELATED WORK

Superposition & latent representations. Research on superposition has identified a variety of feature-interference mechanisms: Nanda (2023) distinguish *representational* from *computational* superposition, while Gurnee et al. (2023) differentiate *alternating* from *simultaneous* interference. Data correlations, for their part, shape the resulting arrangements: pairs of correlated features become orthogonal (Elhage et al., 2022), drive the formation of superposition (Chan, 2024), and give rise to semantic clustering (Prieto et al., 2025a). Beyond superposition, representation geometry is further shaped by spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023), and the optimisation objectives themselves (Casper, 2023). Sparse autoencoders (SAEs), moreover, have been brought to bear upon vision models in order to disentangle features held in superposition (Lim et al., 2024; Joseph et al., 2025).

Adversarial vulnerability. The literature is organised around an attack–defence dichotomy. Attacks (Szegedy et al., 2014) deploy imperceptible perturbations so as to maximise model error, and encompass FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018). Adversarial robustness, correspondingly, denotes the maintenance of performance under attack. Defences include adversarial training (Madry et al., 2018) and certified training (Wong & Kolter, 2018; Palma et al., 2024), the latter furnishing guarantees against perturbations confined within specified input bounds.

The role of input & latent features on adversarial robustness. Vulnerability has been ascribed to predictive yet brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019), as well as to learning shortcuts (Li et al., 2023), while perturbation characteristics are known to vary by dataset (Maiya et al., 2021). Elhage et al. (2022) conjecture a link between superposition and AExs, a proposition contested by Casper (2023). A tension between representational efficiency and adversarial robustness has likewise been observed (Barsbey et al., 2025). Zhang et al. (2021) demonstrate that adversarial perturbations displace latent representations across decision boundaries, and Ganeshan et al. (2019) find that PGD preferentially targets the final layers, proposing feature-based attacks in consequence. Transferability, finally, is constrained by representational discrepancies between models (Li et al., 2023; Wang et al., 2024); the decorrelation of features diminishes transfer (Wiedeman & Wang, 2022), in concordance with our own findings.

7 DISCUSSION & CONCLUDING REMARKS

We have demonstrated that adversarial attacks can exploit the interference patterns that arise from the geometry of superposed features in NNs. Our experiments, spanning both toy models and ViTs, establish that data properties—namely correlations and sparsity—induce distinct superposition geometries, which in turn engender predictable adversarial vulnerabilities. We show, moreover, that these geometric arrangements and the interference they occasion afford the prediction of phenomena such as attack transferability and class-specific susceptibility. Our perspective thus reframes adversarial vulnerability as a potential and inherent consequence of the very manner in which networks efficiently encode vast quantities of information by means of superposition.

Limitations & future work. Our insights derive principally from simplified models. Although our analyses supply an initial mechanistic lens through which to view attacks in large models, the extensive adversarial-robustness literature suggests that the precise mechanisms of vulnerability are apt to vary with attack and model type. Constrained attack types (*e.g.* universal perturbations, jailbreaks, patch attacks), for instance, may prove unable to activate the specific combinations of features required to exploit superposition interference.

Avenues for future work include quantifying the ‘adversarial cost’ of superposition, elucidating its connection to accuracy–robustness trade-offs, and characterising how robust training reshapes the superposition geometry. Since our insights derive from simplified settings in which class features reside in engineered superposition—whereas real networks entail interference among unknown, unlabelled features distributed across multiple layers—we intend to employ SAEs (Bricken et al., 2023) to confront these challenges in large-scale networks, the preliminary results reported in Sec. F indicating this to be a promising direction.

ETHICS STATEMENT

By virtue of its investigative character and generality, our mechanistic interpretability framework is not anticipated to occasion any specific negative impact or immediate societal consequence. Our experiments draw solely upon publicly available datasets (CIFAR-10) and synthetic toy models, and involve neither human subjects nor personally identifiable information. The gradient-based attack methods employed throughout the paper are already well established within the literature; our work introduces no novel attack capability, but rather furnishes a theoretical understanding of pre-existing phenomena. We have no conflicts of interest to declare. In elucidating the relationship between superposition and adversarial vulnerability, we aspire to inform future research directions capable of reconciling model capability with robustness, and thereby to contribute to more trustworthy AI systems that may be safely deployed in critical applications.

REPRODUCIBILITY STATEMENT

We furnish source code for all experiments within the supplementary materials. The toy-model experiments (Sec. 3) are accompanied by complete architectural details, training procedures, evaluation metrics, and data-generation processes. For the ViT experiments (Sec. 4), the architecture and training procedures are provided, with further particulars in Sec. D.1. The modular-arithmetic experiments (Sec. 5) detail the network architecture, the data encoding, and the learned trigonometric algorithm, supplemented by additional details in Sec. E.2. Experiments were conducted on an AMD Ryzen Threadripper 3970X 32-Core with 256GB RAM and an RTX 3090. Additional ResNet-18 and SAE experiments are documented in Appendices Sec. D.2.1 and Sec. F, respectively.

LLM USAGE STATEMENT

This work made use of LLMs to assist with the preliminary literature search, to automate the generation of LaTeX tables from CSV files (all numerical values having been verified), and to suggest alternative phrasings in the interest of manuscript clarity. All substantive research contributions and conclusions remain the original work of the authors.

REFERENCES

- Melih Barsbey, Antônio H Ribeiro, Umut Şimşekli, and Tolga Birdal. On the interaction of compressibility and adversarial robustness. *arXiv preprint arXiv:2507.17725*, 2025.
- Leonard Bereska and Stratis Gavves. Mechanistic interpretability for AI safety - a review. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- Sébastien Bubeck and Mark Sellke. A universal law of robustness via isoperimetry. In Marc’Aurelio Ranzato, Alina Beygelzimer, Yann N. Dauphin, Percy Liang, and Jennifer Wortman Vaughan (eds.), *Advances in Neural Information Processing Systems*, pp. 28811–28822, 2021.
- Stephen Casper. EIS IX: Interpretability and adversaries. *AI Alignment Forum*, February 2023.
- Lawrence Chan. Superposition is not “just” neuron polysemanticity. *AI Alignment Forum*, April 2024.
- Joana C Costa, Tiago Roxo, Hugo Proença, and Pedro Ricardo Morais Inacio. How deep learning sees the world: A survey on adversarial attacks & defenses. *IEEE Access*, 12:61113–61136, 2024.

-
- Valérie Costa, Thomas Fel, Ekdeep Singh Lubana, Bahareh Tolooshams, and Demba Ba. From flat to hierarchical: Extracting sparse representations with matching pursuit. *arXiv preprint arXiv:2506.03093*, 2025.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255, 2009.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, G Heigold, S Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2020.
- Rüdiger Ehlers. Formal verification of piece-wise linear feed-forward neural networks. In Deepak D’Souza and K. Narayan Kumar (eds.), *Automated Technology for Verification and Analysis - 15th International Symposium, ATVA 2017, Pune, India, October 3-6, 2017, Proceedings*, volume 10482 of *Lecture Notes in Computer Science*, pp. 269–286. Springer, 2017.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy models of superposition. *arXiv preprint arXiv:2209.10652*, 2022.
- Alhussein Fawzi, Seyed-Mohsen Moosavi-Dezfooli, and Pascal Frossard. Robustness of classifiers: from adversarial to random noise. *Advances in neural information processing systems*, 29, 2016.
- Aditya Ganeshan, Vivek BS, and R Venkatesh Babu. FDA: Feature disruptive attack. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 8069–8079, 2019.
- Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*, 2025.
- Gabriel Goh. A discussion of ‘adversarial examples are not bugs, they are features’: Two examples of useful, non-robust features. *Distill*, 2019.
- Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*, 2014.
- Clément Guerner, Anej Svete, Tianyu Liu, Alexander Warstadt, and Ryan Cotterell. A geometric notion of causal probing. *arXiv preprint arXiv:2307.15054*, 2023.
- Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. Finding neurons in a haystack: Case studies with sparse probing. *Trans. Mach. Learn. Res.*, 2023.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770–778, 2016.
- Matthias Hein and Maksym Andriushchenko. Formal guarantees on the robustness of a classifier against adversarial manipulation. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Patrick Henriksen and Alessio Lomuscio. Robust training of neural networks against bias field perturbations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(12):14865–14873, Jun. 2023.
- Nikolaus Howe, Ian McKenzie, Oskar Hollinsworth, Michał Zajac, Tom Tseng, Aaron Tucker, Pierre-Luc Bacon, and Adam Gleave. Scaling trends in language model robustness, 2025.
- A. Ilyas, S. Santurkar, D. Tsipras, L. Engstrom, B. Tran, and A. Madry. Adversarial examples are not bugs, they are features. In *Advances in Neural Information Processing Systems 32*, pp. 125–136, 2019.

-
- Sonia Joseph, Praneet Suresh, Lorenz Hufe, Edward Stevinson, Robert Graham, Yash Vadi, Danilo Bzdok, Sebastian Lapuschkin, Lee Sharkey, and Blake Aaron Richards. Prisma: An open source toolkit for mechanistic interpretability in vision and video, 2025.
- Guy Katz, Clark W. Barrett, David L. Dill, Kyle Julian, and Mykel J. Kochenderfer. Reluplex: An efficient SMT solver for verifying deep neural networks. In Rupak Majumdar and Viktor Kuncak (eds.), *Computer Aided Verification - 29th International Conference*, volume 10426, pp. 97–117. Springer, 2017.
- Vignesh Kothapalli. Neural collapse: A review on modelling principles and generalization. *Trans. Mach. Learn. Res.*, 2023.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. CIFAR-10 dataset.
- Victor Lecomte, Kushal Thaman, Rylan Schaeffer, Naomi Bashkansky, Trevor Chow, and Sanmi Koyejo. What causes polysemanticity? an alternative origin story of mixed selectivity from incidental causes. *arXiv preprint arXiv:2312.03096*, 2023.
- Ang Li, Yifei Wang, Yiwen Guo, and Yisen Wang. Adversarial examples are not real features. *Advances in Neural Information Processing Systems*, 2023.
- Hyesu Lim, Jinho Choi, Jaegul Choo, and Steffen Schneider. Sparse autoencoders reveal selective remapping of visual concepts during adaptation. *arXiv preprint arXiv:2412.05276*, 2024.
- Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermy, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language model. *Transformer Circuits Thread*, 2025.
- Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards Deep Learning Models Resistant to Adversarial Attacks. In *International Conference on Learning Representations*, February 2018.
- Shishira R. Maiya, Max Ehrlich, Vatsal Agarwal, Ser-Nam Lim, Tom Goldstein, and Abhinav Shrivastava. A Frequency Perspective of Adversarial Robustness. *arXiv*, 2021.
- Mark Niklas Müller, Franziska Eckert, Marc Fischer, and Martin T. Vechev. Certified training: Small boxes are all you need. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- Preetum Nakkiran. A discussion of ‘adversarial examples are not bugs, they are features’: Adversarial examples are just bugs, too. *Distill*, 2019.
- Neel Nanda. A comprehensive mechanistic interpretability explainer and glossary, 2023.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- Alessandro De Palma, Rudy Bunel, Krishnamurthy Dvijotham, M. Pawan Kumar, and Robert Stanforth. IBP regularization for verified adversarial robustness via branch-and-bound. *CoRR*, abs/2206.14772, 2022.
- Alessandro De Palma, Rudy Bunel, Krishnamurthy (Dj) Dvijotham, M. Pawan Kumar, Robert Stanforth, and Alessio Lomuscio. Expressive losses for verified robustness via convex combinations. In *The Twelfth International Conference on Learning Representations, ICLR, 2024*.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024.

-
- L Prieto, E Stevinson, M Barsbey, T Birdal, and P Mediano. Correlations in the data lead to semantically rich feature geometry under superposition. *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025a.
- Lucas Prieto, Melih Barsbey, Pedro A. M. Mediano, and Tolga Birdal. Grokking at the edge of numerical stability. In *The Thirteenth International Conference on Learning Representations, ICLR*, 2025b.
- Aditi Raghunathan, Jacob Steinhardt, and Percy Liang. Certified defenses against adversarial examples. In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net, 2018.
- Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred A. Hamprecht, Yoshua Bengio, and Aaron C. Courville. On the spectral bias of neural networks. In *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, volume 97 of *Proceedings of Machine Learning Research*, pp. 5301–5310. PMLR, 2019.
- Shashata Sawmya, Linghao Kong, Ilia Markov, Dan Alistarh, and Nir N Shavit. Wasserstein distances, neuronal entanglement, and sparsity. In *The Thirteenth International Conference on Learning Representations*, 2025.
- Adam Scherlis, Kshitij Sachan, Adam S Jermyn, Joe Benton, and Buck Shlegeris. Polysemanticity and capacity in neural networks. *arXiv preprint arXiv:2210.01892*, 2022.
- Ludwig Schmidt, Shibani Santurkar, Dimitris Tsipras, Kunal Talwar, and Aleksander Madry. Adversarially robust generalization requires more data. In *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems 2018, NeurIPS 2018, December 3-8, 2018, Montréal, Canada*, pp. 5019–5031, 2018.
- C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus. Intriguing properties of neural networks. In *2nd International Conference on Learning Representations, ICLR 2014*, 2014.
- Tomasz Tkocz. An upper bound for spherical caps. *The American Mathematical Monthly*, 119(7): 606–607, 2012.
- Donghua Wang, Wen Yao, Tingsong Jiang, Xiaohu Zheng, Junqi Wu, and Xiaoqian Chen. Improving the Transferability of Adversarial Examples by Feature Augmentation. *arXiv*, 2024.
- Christopher Wiedeman and Ge Wang. Disrupting adversarial transferability in deep neural networks. *Patterns*, 3(5), 2022.
- Eric Wong and Zico Kolter. Provable defenses against adversarial examples via the convex outer adversarial polytope. In *International conference on machine learning*. PMLR, 2018.
- Huan Zhang, Hongge Chen, Chaowei Xiao, Sven Gowal, Robert Stanforth, Bo Li, Duane S. Boning, and Cho-Jui Hsieh. Towards stable and efficient training of verifiably robust neural networks. In *8th International Conference on Learning Representations, ICLR*, 2020.
- Shufei Zhang, Zhuang Qian, Kaizhu Huang, Qiufeng Wang, Rui Zhang, and Xinping Yi. Towards better robust generalization with shift consistency regularization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pp. 12524–12534. PMLR, 2021.

APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We employ PGD to generate AExs, an iterative procedure for the construction of attacks (Madry et al., 2018). In the untargeted case it maximises the classifier’s loss function $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$, whereas in the targeted case it minimises $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$, in either instance by means of the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_S \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

(+ for maximisation, – for minimisation). $\mathbf{x}'^{(0)}$ is an initial perturbed input, α is step size, $\mathbf{g}_k$ is the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ (y_{class} being y_{true} or y_{target}), and Π_S projects onto the ϵ -ball $\mathcal{S} = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$, with input domain clipping. We use the ℓ_p norms:

1. ℓ_∞ : Constraint $\|\delta\|_{\ell_\infty} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. Π_S clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. Π_S rescales $\delta = \mathbf{x}' - \mathbf{x}$ if $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

In order to elucidate the manner in which adversarial attacks exploit feature representations, we prove that optimal perturbations weight each input dimension according to the degree to which its corresponding feature aligns with the path toward the decision boundary. Our analysis concerns the linear models, taken without ReLU activation functions.

Consider an input $\mathbf{x} \in \mathbb{R}^d$ encoded via $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ into a latent representation $\mathbf{h} \in \mathbb{R}^m$, the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ constituting an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ ($m < d$). For our argmax task, we observe empirically that the trained encoder–decoder pair renders the decoder the transpose of the encoder: $\mathbf{W}_d = \mathbf{W}_e^\top$. It follows that the logit for class j is computed as $z_j = \mathbf{v}_j^\top \mathbf{h}$, where $\mathbf{v}_j$ denotes the j -th column of $\mathbf{W}_e$, and that the predicted class is $\hat{y} = \arg \max_j z_j$. Input perturbations map to latent perturbations according to $\Delta \mathbf{h} = \mathbf{W}_e \delta$, wherein $\delta = (\delta_1, \dots, \delta_d)^\top$ are the perturbation coefficients.

The decision boundary between classes j and k is the set of points at which their logits coincide: $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$, or equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{h} = 0\}$. The vector $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ thus serves as the normal to this pairwise decision boundary, oriented in the direction that maximally augments the margin $z_k - z_j$. We briefly restate the propositions before furnishing the corresponding proofs.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise movement from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^\top \mathbf{n}$$

Proof. To move a sample from being classified as j to being classified as k , we need to maximise the logit margin $z_k - z_j$. Under perturbation δ , the new margin becomes:

$$z'_k - z'_j = \mathbf{v}_k^\top (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^\top (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^\top (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^\top \mathbf{W}_e \delta = \delta^\top \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^\top \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^\top \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

□

Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

This result lays bare the mechanism by which superposition engenders adversarial vulnerability. Each input feature i is perturbed in proportion to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$ —the differential interference between feature i and the class representations. Under superposition, the attendant non-orthogonality entails that even semantically unrelated features possess non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$, thereby giving rise to exploitable interference patterns. The multi-class setting compounds this vulnerability, for with k classes there arise multiple possible pairwise boundaries, each occasioning a distinct interference pattern. Adversarial perturbations capitalise upon these cross-feature dependencies—manipulating features that bear upon the decision through their interference with the class representations. Herein lies the explanation for the vulnerability we observe empirically: attacks succeed not by directly amplifying the target-class features, but by exploiting the web of interference that superposition creates.

Prop. 1 characterises the optimal perturbation direction for movement from class j to k , supplying the gradient that maximises the margin $z_k - z_j$. Where intervening classes exist (that is, where $z_j < z_i < z_k$ for some class i), following this gradient may prompt the model to predict i before ever reaching the target k . Iterative methods such as PGD accommodate this contingency by recomputing gradients at each step—the global attack path thus emerging from a succession of local decisions rather than from a single optimisation. Furthermore, although our analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ on the basis of our empirical observations, the framework extends to arbitrary decoders: in the general case, with decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ possessing rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, which captures the interference between encoder directions and decoder weights.

Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (i.e., $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. □

Prop. 2 serves to explain why attacks transfer between models trained under similar regimes. When models learn feature representations that differ only by an orthogonal transformation—in essence,

the same geometric structure realised in a different orientation—they share identical vulnerability patterns within input space. The orthogonal transformation preserves all inner products between features, thereby maintaining precisely those interference patterns that attacks exploit. This furnishes an explanation for our empirical observation that models trained on data of identical correlation structure exhibit high attack transferability: they discover similar feature geometries up to rotation, and consequently share adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section furnishes supplementary details, extended results, and further discussion pertaining to the toy-model experiments treated in the main paper. We report model accuracies across a broader range of parameters than was presented in the main text, thereby affording insight into how model capacity and data characteristics such as sparsity condition the learning process, together with the circumstances under which feature superposition arises. We subsequently supply additional visual examples corresponding to Fig. 1, illustrating the mechanics of adversarial attacks under a variety of conditions.

C.1 HYPOTHESES TESTING FRAMEWORK

We here state explicitly the hypotheses corresponding to the three research questions posed in Sec. 3.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We test these hypotheses through controlled experiments:

- $H_1(1)$: We measure the input perturbation profile alignment with a class’s latent representation and the latent attack vector.
- $H_1(2)$: We systematically vary input correlations and measure resulting geometries.
- $H_1(3)$: We quantify transferability rates across models with varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

The toy-model experiments presented in the main paper relied predominantly upon low-dimensional settings for the sake of conceptual clarity. In order to characterise the model’s behaviour more broadly, this subsection details the classification accuracies attained by the CE toy model. These results are reported across varying hidden-layer size (h), number of classes (k), number of features, and levels of sparsity (S), so as to illuminate the circumstances under which the models learn to represent features in superposition. The sparsity level denotes the probability that any individual input feature $x_i^{(j)}$ is set to zero, higher values of S signifying greater input sparsity; we report results as a function of the feature density $1 - S$, that is, the probability that a feature is non-zero. Tab. 2 and Tab. 3 together situate the model’s performance limits and its capacity to learn latent representations in superposition.

Table 2: Classification accuracy of the CE toy model at a fixed bottleneck dimension ($m = 2$) across various numbers of classes (k), total input features (features = $k \times 3$), and input feature densities ($1 - S$). These results illustrate the manner in which input sparsity governs the degradation of performance as the number of classes to be superposed within a constrained latent space grows.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model across varying numbers of classes (k), total input features, bottleneck dimensions (m), and input feature sparsity levels (S). This table reports differing numbers of features per class (p).

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 presents the results of the experiments bearing upon the question *Do Input Correlations Determine Latent Feature Geometry?*, spanning a broader range of classes (k) and hidden-dimension sizes (m).

Table 4: Geometric similarity results across correlation types and superposition configurations.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

How does the scaling of networks bear upon adversarial vulnerability? Although empirical evidence intimates that larger models tend toward greater robustness against adversarial attacks, this effect is weak. Under adversarial training, clearer scaling trends do emerge; the improvements, however, remain largely specific to the attack type employed during training rather than conferring robustness of a general kind (Howe et al., 2025).

Two key phenomena are at play in the scaling of models. On the one hand, larger models command greater capacity for the representation of concepts; on the other, there appears to exist a long tail of concepts useful for a larger model to capture in general tasks such as next-token prediction over internet text. It follows that, notwithstanding this augmented capacity, superposition remains prevalent even within frontier models (Lindsey et al., 2025). Corroborating evidence issues from dictionary-learning methods: SAEs demand ever larger dictionaries for larger models (Gao et al., 2025), which suggests that the number of features scales with model size. This in turn suggests that the fundamental tension driving superposition—that models must compress many features into limited dimensions—does not dissipate with scale.

Since both superposition and adversarial vulnerability persist in large-scale models, we hold that our insights remain pertinent across model scales. Understanding how the geometry of superposition evolves with scale—and thereby, perhaps, mitigating vulnerability—constitutes an intriguing avenue for future work.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

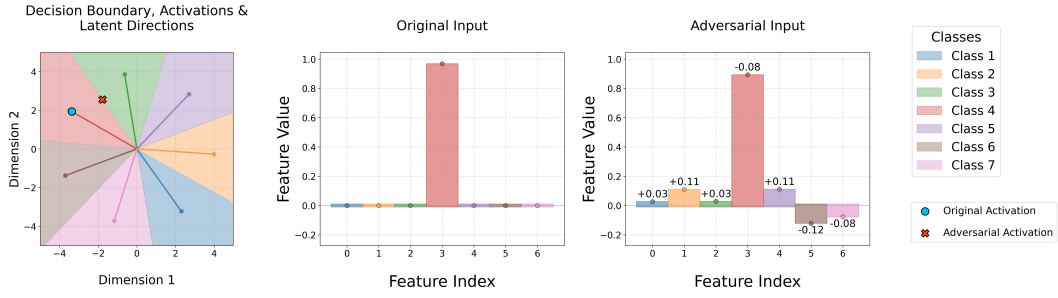
Holding the bottleneck dimension fixed so as to isolate the effects of superposition serves to control for the confounding of superposition pressure with capacity reduction. We fix the bottleneck dimension at $m = 2$ and vary the number of classes k , thereby isolating superposition pressure. We report the model accuracy, robust accuracy, and transferability in Tab. 5. These controls demonstrate that it is heightened superposition pressure (k/m), independently of model capacity, that drives the adversarial vulnerabilities we observe.

C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

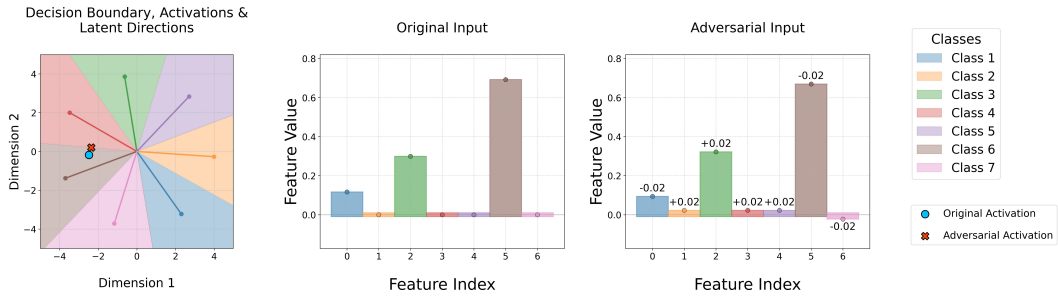
Fig. 1 demonstrates the manner in which adversarial attacks exploit the interference among latent features held in superposition. Here we furnish further visual examples (Fig. 5) so as to reinforce the intuition. Specifically, we supplement the main text by presenting:

Table 5: Performance metrics across correlation types, perturbation budgets (ε), and number of classes (k)

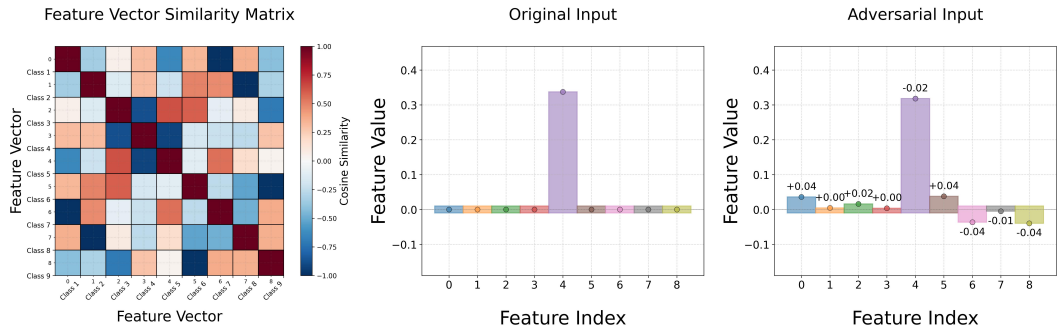
Correlation Type	Parameters		Performance (%)		
	ε	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
Fully Correlated	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0



(a) An ℓ_2 -norm attack changing the classification of an input of class 4 to class 3. The left plot shows original and adversarial activations in latent space, along with representation directions. The right plots show original and perturbed input feature values.



(b) An ℓ_∞ -norm attack changing the classification of an input of class 6 to class 4. The left plot shows original and adversarial activations in latent space relative to class latent directions. The right plots show original and perturbed input feature values.



(c) An ℓ_2 -norm adversarial attack in a 7-class setup with an increased bottleneck dimension $m = 3$. The leftmost plot now shows the cosine similarity matrix between pairs of latent representations for each of the classes.

Figure 5: Visualisations of AExs in the toy model, supplementing Figure 1 of the main paper by illustrating the mechanisms of attack in both activation space and input space under a variety of conditions.

- Fig. 5a shows an additional instance of the setup in Sec. 3.1 ($m = 2, k = 7$) with an ℓ_2 -norm PGD attack, demonstrating the IPP and latent space manipulations that lead to misclassification.
- Fig. 5b shows a similar setup ($m = 2, k = 7$) using an ℓ_∞ -norm PGD attack on a less sparse input.
- Fig. 5c shows an example with increased bottleneck dimensionality ($m = 3, k = 7$) and accompanying ℓ_2 -norm PGD attack. The feature vector similarity matrix used to calculate geometric similarity is also shown.

D CIFAR-10 EXPERIMENTS

In order to ascertain whether the principles observed in the toy models extend to more complex settings, Sec. 4 introduces experiments trained on CIFAR-10 (Krizhevsky, 2009) with an engineered bottleneck. This appendix section furnishes further details concerning this setup and presents extended results.

D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) architecture comprised 6 transformer layers, an embedding dimension (d) of 512, and 8 attention heads within each transformer layer. Input images from the CIFAR-10 dataset, of size 32×32 pixels, were partitioned into patches of 4×4 pixels. The Multilayer Perceptron (MLP) within each transformer block possessed a hidden dimension of 512, and learned positional embeddings were employed.

The bottleneck architecture consisted of a linear encoder succeeded by a linear decoder. The linear encoder projected the pre-classification activations obtained from the ViT backbone into an m -dimensional latent space ($\{2, 3, 5, 10\}$), whereupon the subsequent linear decoder mapped these m -dimensional representations back to the $k = 10$ dimensions corresponding to the CIFAR-10 classes.

The base ViT model was trained on the CIFAR-10 dataset for 250 epochs. A learning rate of 0.001 was adopted in conjunction with the Adam optimiser under the default PyTorch parameters. The batch size was set to 512, and a cosine-annealing learning-rate scheduler was used, with CE as the loss function and dropout applied throughout. Training was conducted across five distinct random seeds so as to account for variability. Once the base ViT model had been trained, its weights were frozen, and the bottleneck layer was thereafter trained for 30 epochs at a learning rate of 0.001. As regards preprocessing, the images undergo RandomCrop with 4-pixel padding, are resized to the target size (32x32 by default), and are subjected to RandomHorizontalFlip for data augmentation.

D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) depicts how normalised robust accuracy varies with bottleneck dimension at a single value of ϵ . This subsection elaborates upon those findings by detailing the normalised robust accuracy under PGD attacks of differing strengths. Results are presented for both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) PGD attacks, thereby affording a more comprehensive view of how the degree of superposition interacts with attack strength to condition model robustness.

The AExs for these evaluations were generated using PGD over 100 iterations at a step size (α) of 0.01. Robust accuracy was evaluated on 500 samples per configuration, and we report the mean normalised robust accuracy together with the standard deviation across five random seeds.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10, subjected to ℓ_∞ -norm PGD attacks of varying perturbation magnitudes (ϵ). Robust accuracy is normalised by the clean accuracy of each bottlenecked model. These results support the findings of Sec. 4, demonstrating a similar trend across ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10, subjected to ℓ_2 -norm PGD attacks of varying perturbation magnitudes (ϵ). Robust accuracy is normalised by the clean accuracy of each bottlenecked model. These results support the findings in Section 4 of the main paper, showing decreasing robustness with smaller m (increased superposition) and larger ϵ .

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

We here report results on attack transferability across a range of perturbation magnitudes (ϵ) and bottleneck dimensions (m). Tab. 9 presents the ℓ_2 -norm attack transferability, and Tab. 8 the ℓ_∞ -norm attack transferability.

D.2 CORRELATIONS IN CLASS FEATURES

In Sec. 3 it was the correlations between inputs that drove the superposition arrangements. We remark that here it is not the correlations between input classes but rather the correlations among the representations at this point in the network that govern these arrangements. At the classification layer, this is likely akin to the confusion matrix—*i.e.* the pattern by which each class is misclassified in relation to the others. Classes that cluster together are precisely those that the network finds inherently similar and misclassifies in concert. To test this proposition, we repeat the experiment, fine-tuning the base ViT using timm/vit_base_patch16_384, which has been trained on ImageNet-21k (14 million images, 21,843 classes) and ImageNet (1 million images, 1,000 classes). After fine-tuning on CIFAR-10 and applying the same bottleneck training procedure as in Sec. 4, the resulting geometry exhibits random ordering across initialisations, with approximately equal spacing between features (*i.e.* neural collapse Kothapalli (2023)). In this case, performance approaches 100%, so that the confusion matrix reduces to the identity and the superposed structure is lost. By contrast, models trained solely on CIFAR-10 converge upon similar geometries precisely because they share the same learned difficulty structure—the same pairs of classes proving challenging to distinguish—which gives rise to consistent superposition patterns.

D.2.1 RESNET-18

In order to address concerns regarding architectural generalisation, we conduct the same experiments using a ResNet-18 (He et al., 2016) architecture as the base model in place of a ViT. We attain a marginally higher clean accuracy of 92% (as against 89% for the ViT). We observe that robust accuracy declines somewhat more rapidly for the ResNet-18 than for the ViT at identical ϵ values. Nevertheless, we observe the same declining trend in normalised robust accuracy and the same rising transferability as bottleneck pressure intensifies.

E MODULAR ADDITION

Sec. 5 introduces a modular-addition task by which to investigate adversarial vulnerability in a scenario wherein features are orthogonal. This section furnishes a detailed experimental setup for this task, together with additional results and further discussion of the manner in which we take this species of analysis to be useful.

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT models. Transferability is shown from a model trained with a specific ‘Source Seed’ (e.g., Seed 10) to three different target models, each trained with one of the seeds listed in the sub-header (e.g., ‘vs. Seeds 20/30/40’). The three slash-separated values in each cell correspond to the transferability to these three target seeds, respectively. All models within a row share the same bottleneck dimension, m . The ‘Mean $\pm$ Std’ column averages transferability across all 12 source-target seed pairings for each (ϵ, m) configuration. This supports the claim in Section 4 that higher superposition can lead to more consistent latent geometries and thus higher transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 $\pm$ 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 $\pm$ 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 $\pm$ 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 $\pm$ 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 $\pm$ 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 $\pm$ 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 $\pm$ 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 $\pm$ 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 $\pm$ 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 $\pm$ 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 $\pm$ 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 $\pm$ 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 $\pm$ 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 $\pm$ 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 $\pm$ 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 $\pm$ 3.3

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As detailed in Nanda et al. (2023) and Prieto et al. (2025b), the network discovers and implements a trigonometric algorithm by which to solve this task: the encoder projects inputs onto non-basis-aligned linear representations that encode $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at the key frequencies $\omega_k = 2\pi k/P$. These features are represented orthogonally—*i.e.* not in superposition—and their sinusoidal values may be read off by way of linear probes. The subsequent MLP thereupon applies sum-angle identities to compute $\cos((a + b)\omega_k)$ for various k , which constructively interfere at the logits so as to select the correct class. The AEs are generated by relaxing the one-hot constraint and perturbing the input $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ via ℓ_2 - and ℓ_∞ -norm PGD. For the sake of simplicity, only a is attacked, while b is held fixed.

As explained in Nanda et al. (2023), the network learns a structured trigonometric algorithm. W_e encodes the sine and cosine values of the integers at the key frequencies; these features are linearly decodable sine/cosine values, and scaling the corresponding feature vector alters the encoded sinusoidal value itself rather than its activation intensity. The integer embeddings $\mathbf{W}_e^j$ project onto a Fourier-like basis, thereby inducing a circular, periodic structure within the embedding space. Unlike the arrangements arising from data correlations, this circularity issues from the sine/cosine encoding of the (uncorrelated) input integers.

Our model learns to exploit approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j predominantly encodes its fourteen corresponding trigonometric features, namely $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$. The concatenated 200-dim. vector thus supplies these ≈ 28 key trigonometric features (fourteen for a and fourteen for b) to the MLP, which leverages them, by way of sum-angle identities (*e.g.* $\cos(A + B) = \cos A \cos B - \sin A \sin B$), to compute

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT models. The table format, detailing source model to target model transferability, mirrors that of Tab. 8; please see its caption for a full explanation. These ℓ_2 results further support the claim in Sec. 4 that higher superposition leads to increased attack transferability.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

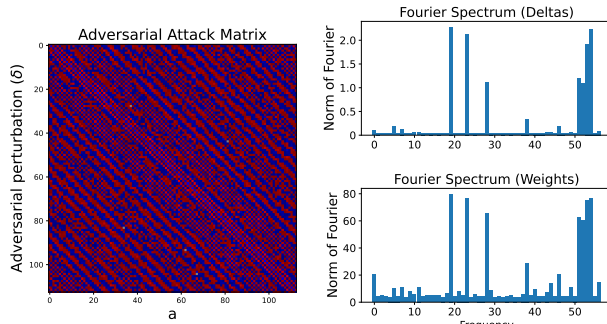


Figure 6: Left: each column depicts the mean attack δ added to a in order to attack $a + b \bmod p$ successfully. Right: the norms of the Fourier components within δ and $W_{MLP_{out}}$.

Table 10: Comparison, across differing attack types, of the minimum ℓ_2 -norm budget required to attack $0 + 0 \bmod p$ successfully.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

$(x_1 + x_2) \bmod P$. The algorithmic brittleness explored in Section 5 stems from the adversarial perturbation of these key learned trigonometric features.

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113. The inputs are integer pairs (a, b) , with $a, b \in \{0, \dots, 112\}$. Each integer is mapped to a 100-dimensional vector by way of a shared embedding $\mathbf{W}_E \in \mathbb{R}^{113 \times 100}$, and the embeddings $\mathbf{W}_E^a$ and $\mathbf{W}_E^b$ are concatenated into a 200-dim. feature vector. This vector feeds a 3-hidden-layer MLP (200 ReLU neurons per layer), the output layer of which provides 113 logits for the classification. The model is trained end-to-end using cross-entropy loss and AdamW (weight decay 4), thereby attaining 100% test accuracy. The AExs are generated using PGD under the ℓ_2 and ℓ_∞ norms with varying ϵ ; the attacks perturb the embedding $\mathbf{W}_E^a$ (with the integer b held fixed) by adding a perturbation δ so as to induce misclassification of $(a + b) \pmod{113}$.

E.3 ALGORITHMIC BRITTLINESS

The frequency-aligned perturbations occasion misclassification by effectively altering the sinusoidal values read off from the encoder features. We invoke the hypothesised neuron-aligned case in order to convey the intuition: $\mathbf{W}_e$ would contain a row corresponding to the sine/cosine for each ω_k , whose element at column $\mathbf{w}_e^a$ holds the value $\sin(\omega_k a)$ or $\cos(\omega_k a)$. For the input a , the one-hot $\mathbf{x}_a$ serves to select $\mathbf{w}_e^a$, yielding $\mathbf{h}_a = \mathbf{w}_e^a$. The perturbation δ then transforms the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, so that $\mathbf{h}'_a = \mathbf{W}_e \mathbf{x}'_a = (1 + \epsilon_a) \mathbf{h}_a + \sum_{k \neq a} \epsilon_k \mathbf{w}_e^k$. The contributions thus introduced by the $k \neq a$ sinusoidal terms corrupt the sinusoidal values, thereby disrupting the MLP’s subsequent computations and precipitating misclassification. Since the model’s algorithm relies upon the constructive interference of each ω_k at the logits, the disruption of any single ω_k does not suffice, the remaining components overwhelmingly supporting the correct answer; effective attacks must accordingly identify a δ that maximally perturbs multiple frequency components at once.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper describes how an understanding of the learned trigonometric algorithm in the modular-addition task permits the analytic construction of adversarial attacks. Sec. 5 of the main paper benchmarks this ‘informed attack’ against PGD and against naive baselines. This section furnishes further details concerning these baseline attack methods. The baselines are as follows:

- **PGD Attack:** Standard PGD used to find the adversary.
- **Informed Attack:** The analytically constructed perturbation, from combinations of cosines.
- **Single a :** This involved finding the minimal attack to a single input to cause misclassification.
- **ℓ_∞ Noise:** This baseline involved adding random noise δ_{noise} to the input a . Each component of δ_{noise} was sampled independently and uniformly from $[-\epsilon, \epsilon]$. The ϵ that successfully induced a misclassification is reported.
- **Uniform δ :** The perturbation vector δ is constrained to have all components equal, i.e., $\delta_j = c \cdot \mathbf{1}$, where c is a scalar. The magnitude of c is reported such that adding δ to the input causes a misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

In order to investigate the impact of defences upon the identified vulnerability, we performed certified robust training, the aim of which was to render the network provably robust against ℓ_∞ -norm perturbations. The certified training method employed was an abstraction-based technique (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022), namely RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023). RSIP-IBP avails itself of Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) and Reversed Symbolic Interval Propagation (RSIP) in order to compute reachable output sets for the network layers, thereby enabling the formal certification of robustness within a given perturbation budget. The training loss function incorporated both a standard loss component and an RSIP loss component. Certified training was conducted over 1000 epochs, the initial 200 of which served as a warm-up phase, at a learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} . The same train/test data splits were employed as in the standard model training. The results of certified training across differing training perturbation budgets (ϵ_{train}), evaluated against various attack perturbation budgets (ϵ_{attack}), are summarised in Tab. 11.

Table 11: Verified robust accuracy (%) for modular addition models under ℓ_∞ -norm perturbations. The table compares a standard (pre-robustification) model with models certifiably trained using RSIP-IBP with different training perturbation budgets (ϵ_{train}). Accuracies are evaluated against various attack budgets (ϵ_{attack}). These results support the discussion in Section 5.1 of the main paper, indicating that certified training improved robustness to specified ϵ_{train} but attacks still succeeded by exploiting the same algorithmic sensitivities at higher ϵ_{attack} .

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

As discussed in Sec. 5, when these robustly trained models were subsequently attacked using PGD, the character of the successful adversarial perturbations remained unchanged: they continued to exploit the very same key frequency components inherent to the model’s learned algorithm. The principal effect of the certified training was thus an increase in the perturbation magnitude required to achieve misclassification for those epsilons against which the model had been successfully certified, rather than any fundamental change in the alignment of the attack vector with the model’s algorithmic sensitivities.

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although the vulnerability explored here arises in the absence of feature superposition, it exhibits a notable phenomenological resemblance to those driven by superposition—namely a shared dependence upon sparsity in the input representation. This resemblance motivates our *reconciliation conjecture*: that algorithmic vulnerabilities may persist in networks possessing orthogonal representations whenever sparsity assumptions are violated upstream. Should the earlier layers of a network produce noisy or non-sparse features—potentially by virtue of adversarial interference—then these deviations may propagate into otherwise well-isolated downstream circuits. In such cases, circuits that operate correctly upon clean inputs may nonetheless fail, for the input they receive is effectively already perturbed. This furnishes a plausible route by which algorithm-disruption mechanisms, such as the frequency attacks studied here, could be activated even in the absence of superposition. Moreover, this interaction between upstream superposition and downstream algorithmic brittleness—and, more broadly, a layer-by-layer understanding of the specific vulnerability mechanisms triggered by AExs—remains underexplored. We consign a detailed investigation, grounded in both empirical evidence and theoretical modelling, to future work as a promising direction.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The analyses of the main paper concentrated upon scenarios featuring either explicitly defined class features (the toy model) or class-level representations within a bottleneck (the ViT experiments). In large, practical networks, however, interference will arise among unlabelled latent features distributed across all layers. As discussed in the future-work section of the main paper, SAEs constitute a promising unsupervised means of extracting more linear features from NNs (Bricken et al., 2023). This appendix section presents a preliminary investigation into the use of SAE-extracted features to characterise how a large-scale ViT responds internally in processing AExs as opposed to clean inputs. These initial results are intended to probe the feasibility of this approach for future work aimed at understanding adversarial phenomena at the level of features within large models, rather than to

advance any conclusive new set of findings. We first outline the experimental setup and validate the behaviour of the SAEs on adversarial inputs, before presenting exploratory layer-wise analyses of feature differences.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of the LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K model, a 12-layer ViT attaining $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009). We employ open-source Prisma SAEs that map the 768-dimensional activation space to a dictionary of 49,152 features (a $64\times$ expansion) (Joseph et al., 2025). We examine SAEs trained upon either mean patch-token representations or CLS-token representations, across all layers, and analyse features activating above a high threshold (0.1)—so as to capture changes to the most highly activating features—as well as above a low threshold (0.001). We generate 10,000 original/successfully-attacked image pairs (ℓ_∞ -norm, $\epsilon = 0.02$) for our SAE feature comparisons. This setup permits an initial exploration of the manner in which adversarial perturbations might manifest at the level of SAE features.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Given that AExs constitute out-of-distribution inputs, and that our aim is to employ a (standardly trained) SAE in order to understand them, we first undertake a basic validation. This subsection examines the impact upon model accuracy and SAE reconstruction when SAEs are inserted at inference time for both clean and adversarial inputs. Our purpose is to establish whether the SAEs process features relevant to the classification task, even under adversarial conditions, prior to any attempt to interpret differences in feature activations.

Table 12: Impact upon ImageNet-1k classification accuracy when a pre-trained SAE is inserted (one layer at a time) during inference for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Accuracy’ denotes the ViT’s top-1 accuracy. ‘Original input + SAE’ signifies that the clean ViT activations at a given layer are passed through the SAE, whereupon its reconstruction is passed to the next layer. The slight ‘de-attacking’ effect (the improved accuracy of ‘Adv. input + SAE’ relative to ‘Adv. input’) intimates that SAEs may filter some adversarial noise; overall, however, the model still largely misclassifies, which indicates that the salient adversarial changes responsible for misclassification are indeed processed.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection presents initial exploratory visualisations comparing SAE feature activations across the several layers of the ViT for original, ℓ_∞ adversarial, and random noisy inputs. These visualisations are intended to bring to light potential avenues for future, more searching investigation into the manner in which adversarial attacks distinctively alter sparse feature representations. We examine the following:

- Activation count: Histograms comparing the number of SAE features activated above a set threshold in a layer for original, adversarial, and noisy images. This explores whether attacks characteristically alter feature sparsity.
- Feature overlap: The degree of overlap (number of common activated features) between (original vs. adversarial) and (original vs. noisy) conditions. This explores if attacks activate a distinctly different set of features.
- Jaccard Similarity: The Jaccard similarity between sets of activated features, offering a normalised measure of overlap.
- Mean activation: Histograms of mean activation values for features active above a threshold. This looks for changes in the intensity of (key) features.
- Distinct Features: The count of unique features activated across an image set for each condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection presents a preliminary analysis of SAE features derived from CLS tokens, considering only those features activated above a threshold of 0.1. Figures 7 through 10 visualise and compare, respectively, the histogram of activation counts, the feature overlap, the Jaccard similarity, and the histogram of mean activations, for original, adversarial, and random noisy inputs across the several ViT layers.

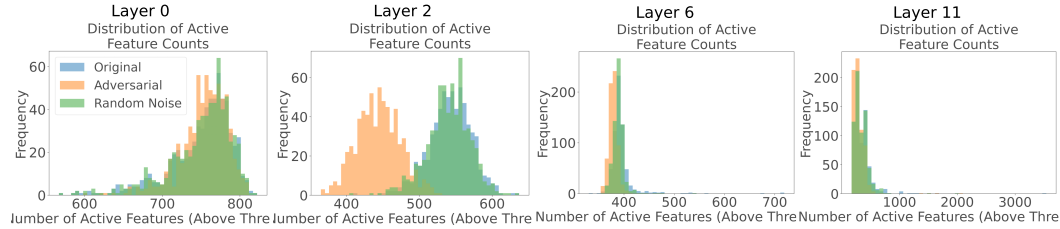


Figure 7: Histogram of activation counts for CLS token SAE features (threshold 0.1).

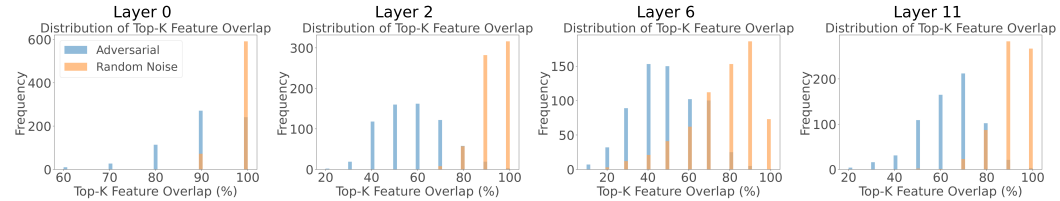


Figure 8: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.1).

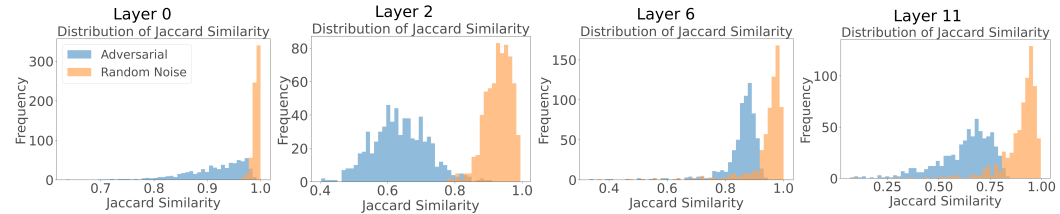


Figure 9: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.1).

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Here we examine SAE features from CLS tokens at a lower activation threshold of 0.001, so as to capture more than merely the most intensely activated features. The subsequent figures illustrate the histogram of activation counts (Fig. 11), the feature overlap (Fig. 12), the Jaccard similarity (Fig. 13),

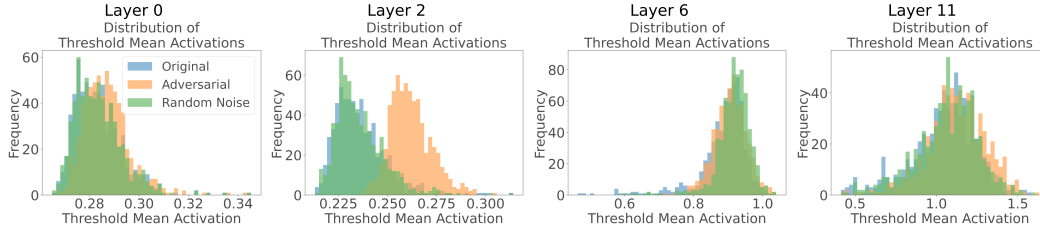


Figure 10: Histogram of mean thresholds for CLS token SAE features (threshold 0.1).

and the histogram of mean activations (Fig. 14) across layers for original, adversarial, and noisy inputs.

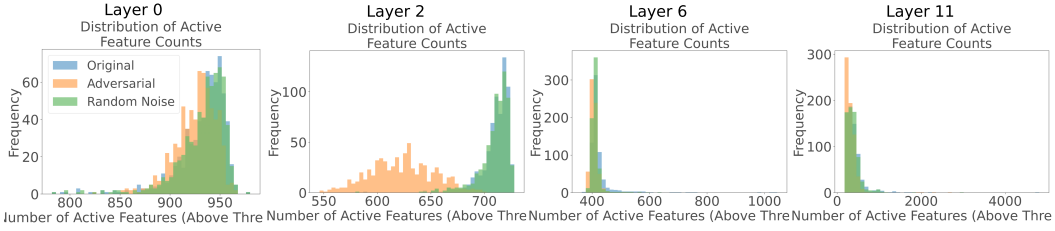


Figure 11: Histogram of activation counts for CLS token SAE features (threshold 0.001), with noise.

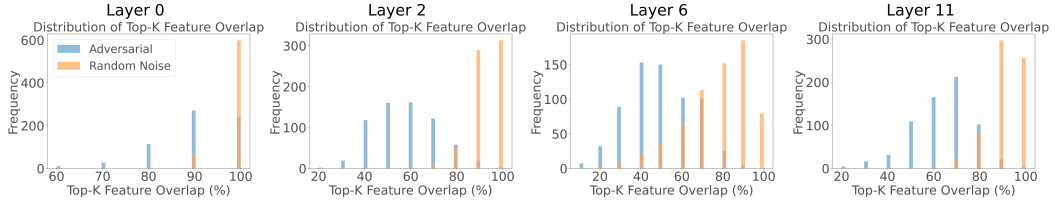


Figure 12: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.001).

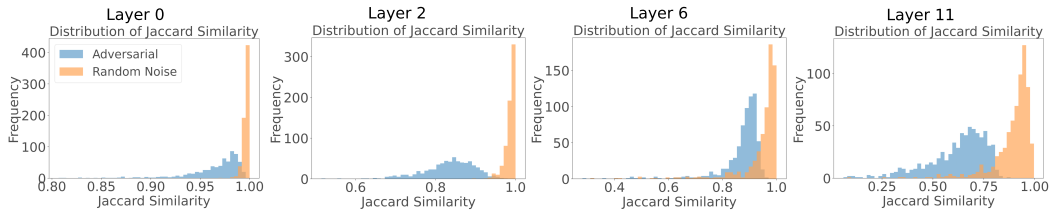


Figure 13: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.001).

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection shifts the focus to SAE features derived from the mean of the patch tokens, again at a high activation threshold of 0.1. We present comparisons of the activation-count histograms (Fig. 15), the distinct-feature-count histograms (Fig. 16), the feature overlap (Fig. 17), the Jaccard similarity (Fig. 18), and the mean-activation histograms (Fig. 19) for original, adversarial, and noisy inputs.

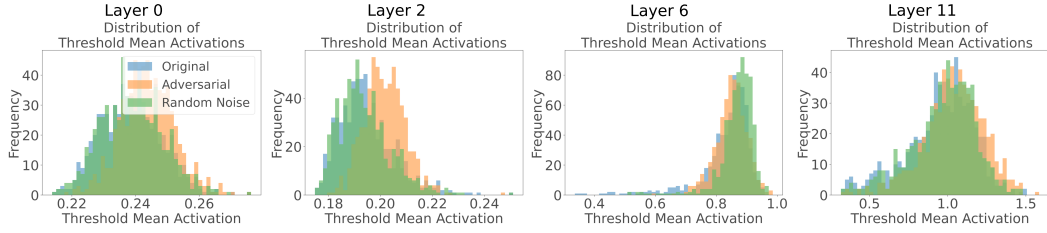


Figure 14: Histogram of mean activations for CLS token SAE features (threshold 0.001), with noise.

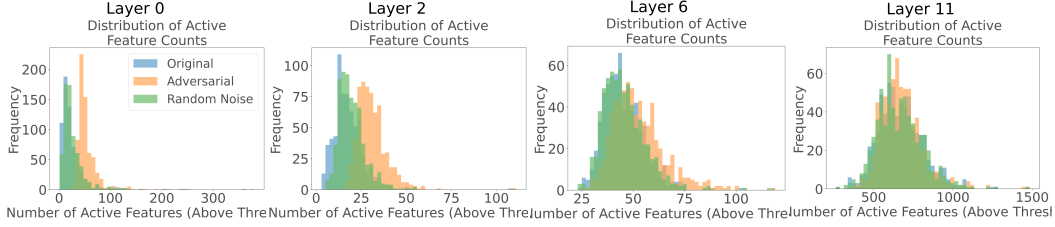


Figure 15: Histogram of activation counts for patch token SAE features (threshold 0.1), with noise.

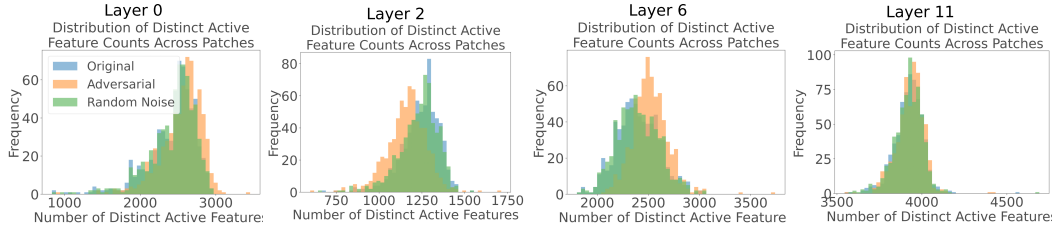


Figure 16: Histogram of distinct patch features for patch token SAE features (threshold 0.1), with noise.

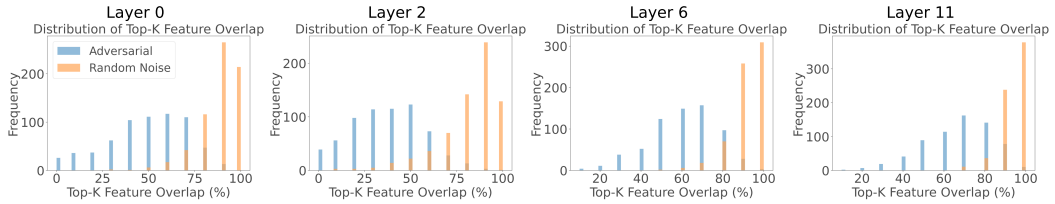


Figure 17: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.1).

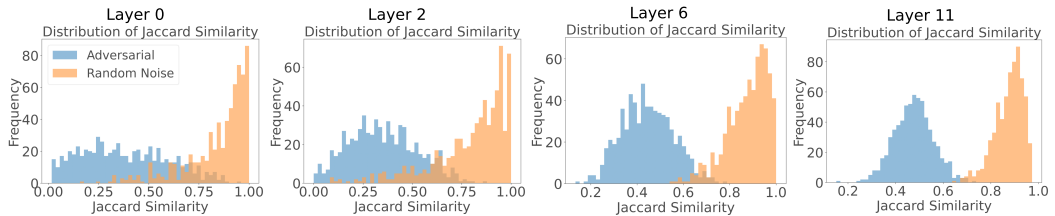


Figure 18: Comparison of Jaccard similarity for patch token SAE features (threshold 0.1).

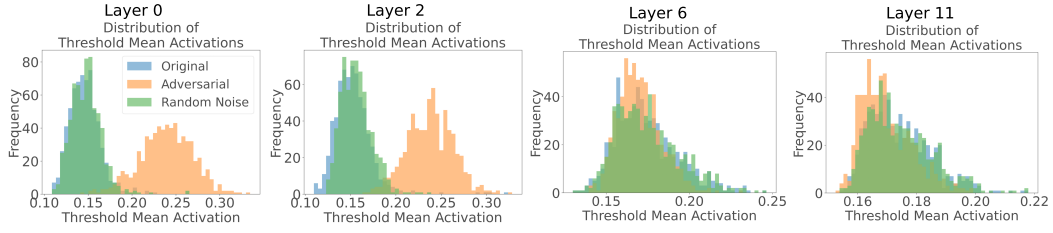


Figure 19: Histogram of mean thresholds for patch token SAE features (threshold 0.1), with noise.

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, this subsection explores SAE features from the patch-token means at a low activation threshold of 0.001. The figures display the histograms of activation counts (Fig. 20), the distinct-feature counts (Fig. 21), the feature overlap (Fig. 22), the Jaccard similarity (Fig. 23), and the mean-activation histograms (Fig. 24) for original, adversarial, and noisy inputs.

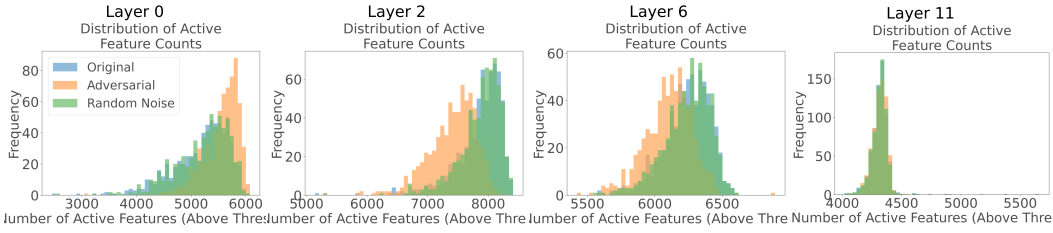


Figure 20: Histogram of activation counts for patch token SAE features (threshold 0.001), with noise.

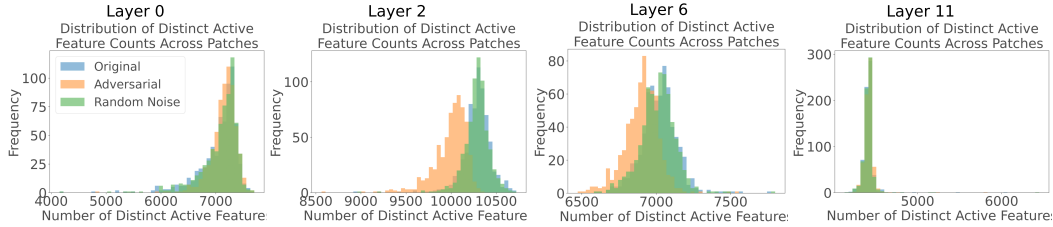


Figure 21: Histogram of distinct patch features for patch token SAE features (threshold 0.001), with noise.

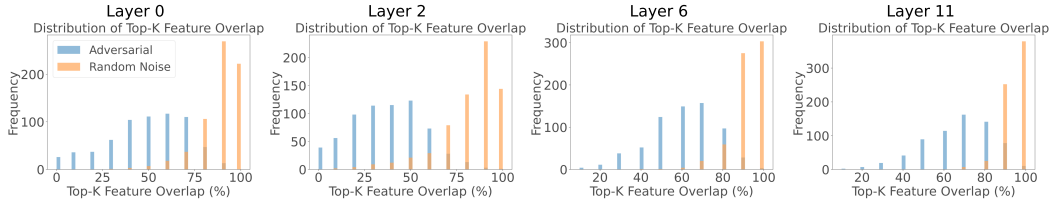


Figure 22: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.001).

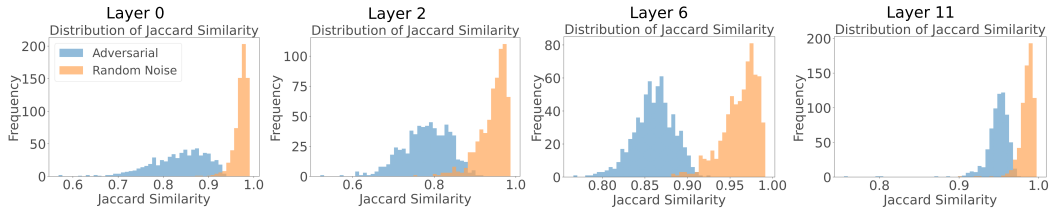


Figure 23: Comparison of Jaccard similarity for patch token SAE features (threshold 0.001).

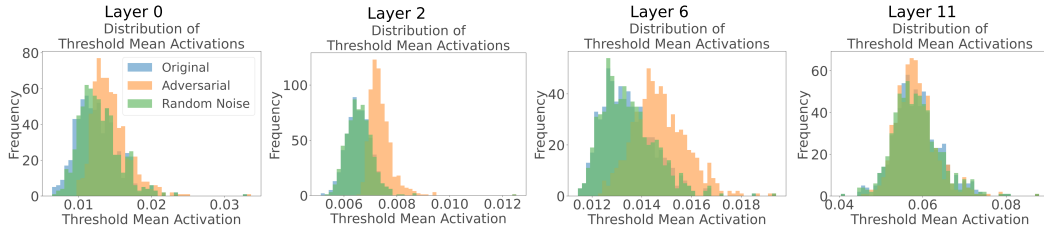


Figure 24: Histogram of mean activations for patch token SAE features (threshold 0.001), with noise.

G ACRONYMS

AEx adversarial example

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

ViT vision transformer